

# PEDIATRIC GASTROENTEROLOGY

## Abdominal pain — pathophysiology (3 categories)

- **Visceral (splanchnic) pain:** visceral fibres **bilateral + unmyelinated**, enter spinal cord at multiple levels → **dull, poorly localized, midline**. Triggered by ischaemia, inflammation, stretch of peritoneum.
- **Parietal (somatic) pain:** stimulation of **parietal peritoneum** → **sharp, intense, discrete, localized**; worse on coughing/movement. Transmitted via myelinated fibres to specific dorsal root ganglia (same side + dermatome).
- **Referred pain:** felt at site distant from organ (shared neuronal pathways). Classic = **pneumonia presents with abdominal pain** (T9 dermatome shared by lung + abdomen).

## Acute abdominal pain in children

- **Most common medical cause = gastroenteritis** (viruses: rotavirus, norovirus, adenovirus, enterovirus; bacteria: E. coli, Yersinia, Campylobacter, Salmonella, Shigella).
- **Causes by age/type:**
- **Acute, child <2 yrs:** surgical (**malrotation, volvulus, intussusception, appendicitis, necrotising enterocolitis, incarcerated inguinal hernia**); medical (GE, UTI, hepatitis, pneumonia, bacterial peritonitis).
- **Chronic, child >2 yrs:** surgical (appendicitis, Meckel's diverticulum, peritonitis, cholecystitis); medical (GE, UTI, lead poisoning, **non-specific abdominal pain, IBD, mesenteric lymphadenitis**).
- **DDx list:** infantile colic, constipation, mesenteric adenitis, trauma, intestinal obstruction, PID, UTI, Meckel's.
- **"The pain": use analgesia as required — it does NOT affect diagnostic accuracy.**

### ### Specific causes

- **Infantile colic:** up to **10% of infants**; first 6 months; scream + draw knees up; paroxysmal crying (evening), relieved by flatus/stool; normal feeding/exam, no associated symptoms.
- **Constipation:** over-diagnosed; can be visceral/LIF; acute (organic) or chronic (functional); diagnosis of last resort.
- **Mesenteric adenitis:** preceding **URTI/adenovirus**; mimics appendicitis (more diffuse pain, higher temperature, **shifting tenderness**, other lymphadenopathy).
- **Abdominal trauma:** accidental vs **non-accidental (parents may conceal info)**.
- **Intestinal obstruction:** malrotation, volvulus, **intussusception, incarcerated hernia, adhesions (acquired or congenital)**.

## Appendicitis (detailed)

- **Types:** acute (inflammatory/suppurative), perforated/complicated (perforated, gangrenous), or normal. Mass → **appendix mass / appendix abscess**.
- **Pathogenesis:** primary luminal obstruction (filled with mucus, swells) → ↑pressure → vessel compression → ischaemia/necrosis (gangrene) → bacterial invasion → **perforation** → **peritonitis/abscess** → **septicaemia**.
- **Symptoms:** **central pain (vague)** → **moves to RIF (localized, sharp, constant, wants to lie still)**; anorexia, nausea, vomiting, mild pyrexia, change in bowel habit (diarrhoea/constipation), urinary symptoms.

- **Signs:** localized tenderness + guarding, **Rovsing's sign**, **psoas sign** (pain on hip extension), **obturator sign** (pain on hip internal rotation), caecal gurgle, **McBurney's point**, rebound tenderness.
- **Appendix positions:** retrocaecal (poor localizing signs), retroileal (diarrhoea), pelvic (diarrhoea/bladder symptoms).
- **Investigations:** bloods (**neutrophil leucocytosis**, **lymphopenia**, ↑**CRP**, sickle test), radiology (AXR — obstruction, faecolith, soft tissue mass, loaded colon; **US** — gynaecological, abscess/mass/thick bowel loops, stones).
- **Treatment:** prep for theatre (analgesia, fluids, antibiotics, consent) → appendicectomy.

## Henoch-Schönlein purpura (HSP)

- **Systemic vasculitis; IgA immune complex deposition in skin + kidney; young children.** Self-limiting, no treatment (symptom control). 1/3 relapse; **irreversible kidney damage ~1%**. Post-viral/bacterial / drug reaction.

## Vomiting in children

- **By cause (colour of vomit):**
- **Obstruction proximal to duodenum** (pyloric stenosis/IHPS, achalasia, web): **non-bilious**.
- **Obstruction distal to duodenum** (congenital/acquired obstruction, foreign body): **bilious**.
- **Reflux/GER:** non-bilious.
- **CNS** (meningitis, encephalitis, head injury, hydrocephalus): non-bilious.
- **Renal** (renal failure, calculi), endocrine/metabolic.

## GER & GERD

- **GER** = physiological reflux. **GERD** = pathologic.
- **Why GER becomes GERD:** aberrance in normal physiology (insufficient clearance/buffering, ↓gastric emptying, abnormal epithelial repair, ↓neural reflexes) → **erosive oesophagitis** → **oesophageal shortening** → **hiatal herniation**.
- **Diagnostic approach:** symptoms/signs, history/exam, **upper GI series**, **oesophageal pH monitoring** (detects acid reflux + temporal symptom association; can't detect non-acid reflux or complications), endoscopy + biopsy, empirical medical therapy.
- **Conservative therapy:**
- **Infants:** normalize feeding volume/frequency, **thickened formula**, non-prone positioning during sleep, hypoallergenic formula trial.
- **Older children:** avoid large meals, don't lie down after eating, lose weight, avoid caffeine/chocolate/spicy, **avoid tobacco smoke**.

## Constipation

- **Definition:** hard stool passed with difficulty **every 3 days, persisting ≥2 weeks**.
- **Organic causes:** structural (anal stenosis, fissure), neuromuscular (**Hirschsprung disease**, **spina bifida**), metabolic (hypokalaemia, hypercalcaemia, **hypothyroidism**, DM), drugs (narcotics, anticholinergic, lead), intestinal (IBD, coeliac, CF, tumours).
- **Functional:** healthy child; painful defecation episode → withholding; toilet training; school toilet avoidance. **Stool-withholding** (stiffen legs, squat, push, cry — misinterpreted as straining). + abdominal pain, distension, flatulence.
- **Clinical:** rectal bleeding, poor appetite, enuresis, UTI; **soiling (overflow incontinence)** in long-standing. Exam: distension, **palpable faecal masses**, DRE (sphincter, ampulla).
- **Functional diagnosis:** favourable response to treatment supports it; refractory → **barium enema + anorectal manometry** (rule out organic, e.g. Hirschsprung).

- **Bristol stool chart:** Type 1 (separate hard lumps — very constipated) → Type 7 (liquid — inflammation). Types 3–4 normal.
- **Treatment (functional):**
  1. **Patient education + regular bowel training** (toilet sitting 5–10 min after meals).
  2. **Disimpaction:** enema or polyethylene glycol (PEG).
  3. **Stool softening:** lactulose, high fibre. **Maintenance:** osmotic laxatives (lactulose, PEG/Movicol), stool softeners (docusate), stimulant laxatives, lubricants (mineral oil) until regular pattern established (several months).
- **Squatting = only natural defecation posture** (anorectal angle).

## High-yield one-liners

- **Abdominal pain:** visceral = midline/dull; parietal = localized/sharp; referred (pneumonia → abdomen).
- **Acute abdominal pain commonest medical cause = gastroenteritis; analgesia doesn't affect diagnosis.**
- **Appendicitis:** central → RIF migration; Rovsing's/psoas/obturator; retrocaecal = poor signs.
- **Mesenteric adenitis:** post-URTI, shifting tenderness, mimics appendicitis. HSP = IgA vasculitis, skin + kidney.
- **Vomiting:** bilious = distal obstruction; non-bilious = pyloric/GER/CNS.
- **GER vs GERD; pH monitoring; infants → thickened formula + positioning.**
- **Constipation = hard stool every 3 days  $\geq$  2 weeks; functional commonest; PEG/enema disimpaction; Bristol chart; Hirschsprung if organic suspected.**

# PEDIATRIC GASTROENTEROLOGY — Revision Table

*Bold = high-yield. Dr Heersh Raof Saeed.*

| ABDOMINAL PAIN          |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Type                    | Key points                                                                                           |
| Visceral                | Bilateral unmyelinated; <b>dull, poorly localized, midline</b> ; ischaemia/inflammation/stretch      |
| Parietal (somatic)      | Parietal peritoneum; <b>sharp, intense, localized</b> ; worse on movement/cough                      |
| Referred                | Distant site; <b>pneumonia → abdominal pain (T9)</b>                                                 |
| Acute <2yr              | Surgical: <b>malrotation, volvulus, intussusception, NEC, incarcerated hernia</b> ; medical: GE, UTI |
| Chronic >2yr            | Appendicitis, Meckel's; <b>non-specific pain, IBD, mesenteric lymphadenitis</b>                      |
| Commonest medical cause | <b>Gastroenteritis</b> (rota/noro/adeno; E. coli, Salmonella, Shigella, Campylobacter)               |
| Analgesia               | <b>Use as required — doesn't affect diagnostic accuracy</b>                                          |

| SPECIFIC CAUSES     |                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Condition           | Key points                                                                                                                                 |
| Infantile colic     | <b>Up to 10% infants, &lt;6mo</b> ; scream + draw knees up; evening, relieved by flatus; normal exam                                       |
| Mesenteric adenitis | <b>Post-URTI/adenovirus</b> ; mimics appendicitis; <b>shifting tenderness</b> , higher temp                                                |
| Appendicitis        | <b>Central → RIF pain; Rovsing's, psoas, obturator signs, McBurney's</b> ; neutrophil leucocytosis + lymphopenia; retrocaecal = poor signs |
| HSP                 | <b>IgA vasculitis (skin + kidney)</b> ; young children; self-limiting; kidney damage ~1%                                                   |

| VOMITING & GERD       |                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------|
| Topic                 | Key points                                                                                                 |
| Vomiting colour       | <b>Proximal obstruction (pyloric/IHPS) = non-bilious; distal obstruction = bilious</b> ; CNS = non-bilious |
| GER vs GERD           | GER physiological; <b>GERD pathologic</b> (erosive oesophagitis → hiatal herniation)                       |
| Diagnosis             | <b>pH monitoring</b> (acid reflux + symptom association), UGI series, endoscopy + biopsy                   |
| Conservative (infant) | <b>Thickened formula, non-prone positioning, hypoallergenic trial</b>                                      |
| Conservative (older)  | Avoid large meals/caffeine/spicy, weight loss, <b>no tobacco</b>                                           |

| CONSTIPATION  |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| Topic         | Key points                                                                                                |
| Definition    | <b>Hard stool, difficulty, every 3 days, ≥2 weeks</b>                                                     |
| Organic       | <b>Hirschsprung, spina bifida, hypothyroidism, anal stenosis, IBD, CF, lead</b>                           |
| Functional    | Commonest; painful defecation → <b>stool withholding; soiling (overflow)</b>                              |
| Diagnosis     | Clinical; refractory → <b>barium enema + anorectal manometry</b> (rule out Hirschsprung)                  |
| Bristol chart | Type 1 (hard lumps) → 7 (liquid); <b>3–4 normal</b>                                                       |
| Treatment     | <b>Education + bowel training; disimpaction (enema/PEG); softening (lactulose); maintenance laxatives</b> |